

StreamCatTools: An R package for working with StreamCat and LakeCat watershed data in R

Marc H. Weber¹, Ryan A. Hill², Selia Markley⁴, Travis Hudson³, and Alan Brookes²

¹ Office of Water, United States Environmental Protection Agency ² United States Environmental Protection Agency Retired ³ Oak Ridge Associated Universities Student Services Contractor c/o United States Environmental Protection Agency ⁴ Oak Ridge Institute for Science and Education Fellow c/o United States Environmental Protection Agency

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Summary

StreamCatTools provides functions for easily working with, visualizing and analyzing StreamCat (Hill, Weber, Leibowitz, Olsen, & Thornbrugh, 2016) and LakeCat (Hill, Weber, Debbout, Leibowitz, & Olsen, 2018) watershed metrics within **R**. The StreamCat and LakeCat datasets provide hundreds of landscape metrics for both the local catchment (e.g. landscape draining to a particular stream reach) and full watershed for every stream reach and lake depicted in the medium resolution National Hydrography Dataset Plus Version 2.1 (NHDPlusV21)(McKay et al., 2012) for the contiguous United States (CONUS). **StreamCatTools** functions wrap the application programming interface (API) for the StreamCat and LakeCat online database and facilitate simple, straightforward access and use of these datasets within R.

Statement of Need

Easily accessible, robust, and consistent watershed data is an underpinning of hydrology research, water quality monitoring programs, and predictive modelling applications, to

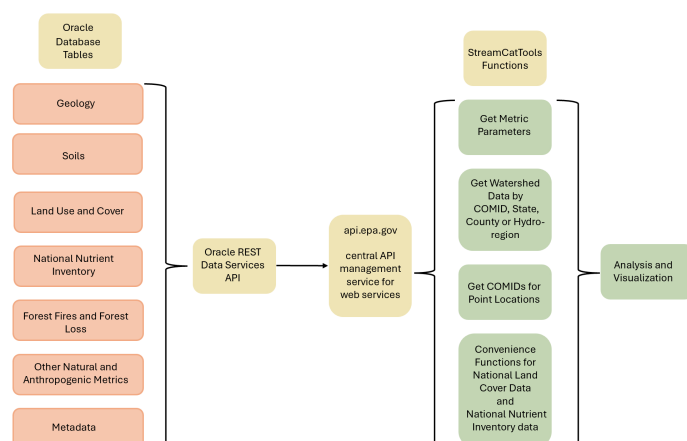


Figure 1: Diagram of the StreamCat and StreamCatTools framework. A backend Oracle database and web service are exposed through api.gov to functions in StreamCatTools which simplify access and analysis of the data in R via the application programming interface.

name just a few examples. The StreamCat (Hill et al., 2016) and LakeCat (Hill et al., 2018) datasets fill this need by providing nationally consistent curated watershed data for CONUS that has had stringent quality control applied. The data encompasses hundreds of watershed metrics for every stream reach and lake feature represented in the National Hydrography Dataset Plus Version 2.1 (NHDPlusV21) (McKay et al., 2012). StreamCatTools fills the need for easily accessible watershed metrics for CONUS by: (1) providing a simple interface in the R programming language to the StreamCat and LakeCat web services, (2) providing convenient functionality to find available StreamCat and LakeCat metric names and information about variables, and (3) extracting StreamCat and LakeCat metrics by COMID (a unique identifier in the NHDPlusV2 framework), by state, by county, by NHD Hydro-region, or for all of CONUS. Providing this valuable watershed data via web services in R follows the FAIR principles laid out in (Wilkinson et al., 2016).

Package Overview

StreamCatTools provides a simple streamlined set of functions to easily query and ingest watershed landscape metrics into an R session. Figure 1 shows the overall framework of the StreamCat database, application programming interface, and functionality in the package that simplifies data access in R using web services for the StreamCat and LakeCat datasets.

The core functions in **StreamCatTools** leverage the **httr2** library (Wickham, 2025) for a modern, pipeable method for working with APIs. Specifically, **StreamCatTools** simplifies the calls to the API for StreamCat and LakeCat data within the R programming language to allow a user to: (1) access details on available StreamCat and LakeCat metrics, (2) match users' sites to NHDPlusV21 waterbodies (streams or lakes) to access StreamCat and Lakecat data, and (3) retrieve StreamCat and LakeCat watershed metrics by COMID, by state, by county, by hydroregion, or for all of CONUS. Additionally, several convenience functions retrieve data from particular years for data with time series, such as the National Land Cover Database (NLCD) (USGS, 2024) and the National Nutrient Inventory (NNI) (Brehob et al., 2025).

You can install the most recent version of **StreamCatTools** from CRAN by running:

```
# install.packages("StreamCatTools")
```

You can install the most recent development version from GitHub by running:

```
# library(remotes)
# install_github("USEPA/StreamCatTools", build_vignettes=FALSE)
```

StreamCatTools is loaded into an **R** session:

```
library(StreamCatTools)
```

StreamCatTools includes several functions to facilitate accessing and working with the data. First, users can list metric names and find out more about StreamCat and LakeCat data available in StreamCat using the `sc_get_params` and `lc_get_params` functions. The parameters `aoi` and `metric_names` provide a user with the available areas of interest for metrics and names of metrics. The areas of interest available for all metrics in StreamCat and Lakecat are 'cat' and 'ws' (catchment and watershed) and in StreamCat some metrics are available at the 100 meter riparian buffer scale ('catrp100' and 'wsrp100') and for certain metrics the designation of 'other' needs to be used (where the metric is in-stream or derived and not part of the other listed areas of interest). Extracting this information in StreamCatTools looks like this:

```
aois <- sc_get_params(param='aoi')
name_params <- sc_get_params(param='metric_names')
print('areas of interest include: ')
```

```
## [1] "areas of interest include: "
```

```
## cat
## catrp100
## other
## ws
## wsrp100
```

```
print('A selection of available StreamCat metrics include: ')
```

```
## [1] "A selection of available StreamCat metrics include: "
```

```
## agkffact
## al2o3
## bankfulldepth
## bankfullwidth
## bfi
## canaldens
## cao
## cbnf
## chem
## clay
```

And the same information for LakeCat metrics is derived in similar fashion:

```
aois <- lc_get_params(param='aoi')

name_params <- lc_get_params(param='metric_names')

## Warning in open.connection(con, "rb"): URL
## 'https://api.epa.gov/StreamCat/lakes/metrics': Timeout of 60 seconds was
## reached

## An error occurred during req_perform(); the service may be down or function par

print('areas of interest include: ')

## [1] "areas of interest include: "

## cat
## ws

print('A selection of available LakeCat metrics include: ')

## [1] "A selection of available LakeCat metrics include: "
```

StreamCat and LakeCat are built around the concepts of local drainage area (i.e catchment) and watershed (i.e. the local drainage area and all upstream catchments) (Hill et al., 2016). This approach uses the NHDPlusV21 hydrographic framework of catchments as the building block for summarizing landscape information represented in landscape data and then programmatically summarizing both the catchment landscape summary and all upstream catchments. `sc_get_params` and `lc_get_params` also include a ‘variable_info’ parameter to return more detailed metadata for metrics including both short and long metric descriptions, years available (if applicable), units, and the metric category.

```
var_info <- sc_get_params(param='variable_info')

my_data <- head(var_info[,c('metric', 'short_description', 'long_description',
  ↪ 'units')], 10)

knitr::kable(
  my_data,
  format = "latex",
  booktabs = TRUE,
  longtable = TRUE,
  col.names = c('metric', 'short_description', 'long_description', 'units'),
  align = "ccccc"
)
```

metric	short_description	
NABD_Dens[AOI]	NABD Dam Density	
NABD_NIDStor[AOI]	NABD NID Reservoir Volume	
NABD_NrmStor[AOI]	NABD Normal Reservoir Volume	
Precip_Minus_EVT[AOI]	Surplus Precipitation	This dataset represents
agkffact[AOI]	Ag Soil Erodibility Kf Factor	Mean of STATSGO Kffactor

al2o3[AOI]	Mean Lithological Aluminum Oxide
areasqkm[AOI]	Watershed Area Square Kilometers
bankfulldepth	Predicted Bankfull Depth
bankfullwidth	Predicted Bankfull Width
bfi[AOI]	Base Flow Index

Bas

Additional functions provided for getting metadata about the underlying StreamCat and LakeCat data include the `sc_fullname` and `lc_fullname` functions and the `sc_get_params` and `lc_get_params` functions, respectively. Users can also filter metric names and information by the metric year(s), the indicator categories for metrics, the metric dataset names, or the areas of interest the metrics are available for using the `sc_get_metric_names` or `lc_get_metric_names` functions. More details on these functions can be found at the [package introduction page](#).

The primary package functionality is in the `sc_get_data` and `lc_get_data` functions which allow users to extract catchment or watershed metrics of interest by providing NHDPlusV21 COMIDs for streams or lakes within the database. Users can also request data for a given state(s), county(ies), hydroregion(s), or all of CONUS. Additionally, convenience functions are provided for accessing the NLCD and NNI datasets, respectively, using `sc_nlcd` and `lc_nlcd` and `sc_nni` and `lc_nni`.

The following example shows a request for two catchment and watershed metrics for three stream reaches:

```
df <- sc_get_data(metric='pcturbmd2019,damdens',
                  aoι='cat,ws',
                  comid='179,1337,1337420')
```

The first line requests percent of NLCD medium intensity developed land cover in 2019 and the density of dams. The second line specifies that the geographic area of interest (aoι) is the drainage to the local reach scale (i.e. 'cat', short for catchment) and the full watershed. Finally, the final line request this data for three NHDPlusV2 stream segments specified by their unique COMIDs.

Applications and Discussion

Watershed metrics from `StreamCatTools` can be easily visualized with functions from `nhdplusTools` (Blodgett & Johnson, 2023) and `ggplot2` (Wickham, 2016). The example below plots the NLCD percent imperviousness for the the local drainage (catchment in NHDPlusV2 syntax) and displays the values mapped to each stream reach and to the overall basin boundary:

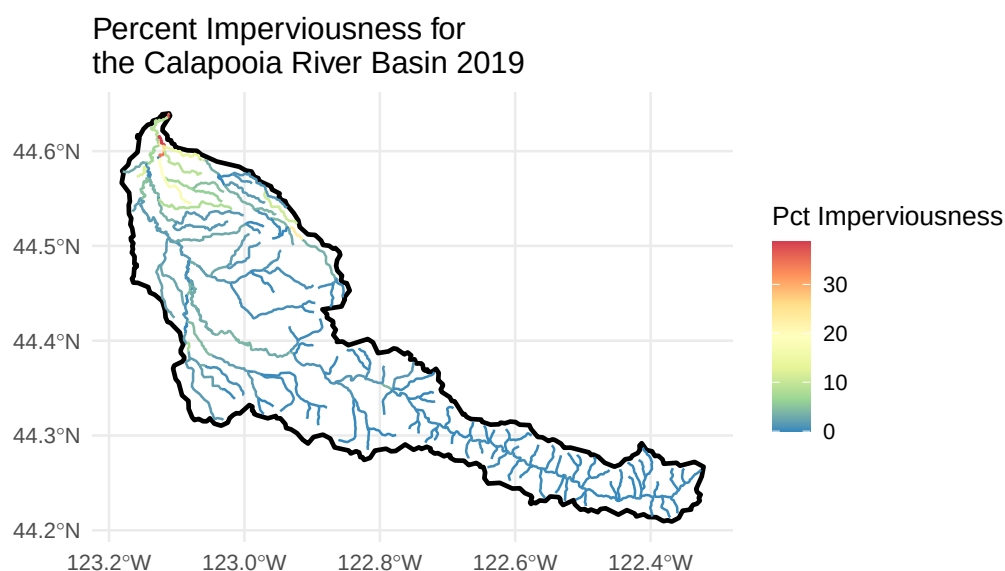
```
library(nhdplusTools)
library(ggplot2)
library(ggspatial)
library(StreamCatTools)

start_comid = 23763517
nldi_feature <- list(featureSource = "comid", featureID = start_comid)
flowline_nldi <- nhdplusTools::navigate_nldi(nldi_feature, mode = "UT",
  ↳ data_source = "flowlines", distance=5000)
df <- sc_get_data(metric='pctimp2019', aoι='cat',
  ↳ comid=flowline_nldi$UT_flowlines$nhdplus_comid)
flowline_nldi <- flowline_nldi$UT_flowlines
```

```
flowline_nldi$PCTIMP2019 <- df$pctimp2019cat[match(flowline_nldi$nhdplus_comid,
  ↪ df$comid)]
basin <- nhdplusTools::get_nldi_basin(nldi_feature = nldi_feature)

calapooia <- ggplot() +
  geom_sf(data = basin,
    fill = NA,
    color = "black",
    linewidth = 1) +
  geom_sf(data = flowline_nldi,
    aes(colour = PCTIMP2019)) +
  scale_y_continuous() +
  scale_color_distiller(palette = "Spectral") +
  labs(color = "Pct Imperviousness") +
  theme_minimal(12) +
  ggtitle('Percent Imperviousness for \nthe Calapooia River Basin 2019')

plot(calapooia)
```



Also show: new functionality for accessing and plotting NNI data. Perhaps any other uses or applications we are aware of (such as CASTools R Shiny app).

Acknowledgements

Examples of using StreamCat and LakeCat make extensive use of `nhdplusTools` (Blodgett & Johnson, 2023) and the functions for accessing the API are facilitated through use of `httr2` (Wickham, 2025). Figures were created using `ggplot2` (Wickham, 2016).

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